

Computer Science & Information Technology

Computer Organization & Architecture

DPP: 1

Pipeline Processing

Q1 Consider the given set of instructions, which are having instruction format as follows:

Opcode, Destination operand, Source1, Source2

- | | |
|-------------------|-------------------------|
| 1. ADD R2, R1, R0 | $R2 \leftarrow R1 + R0$ |
| 2. MUL R4, R3, R2 | $R4 \leftarrow R3 * R2$ |
| 3. SUB R6, R5, R4 | $R6 \leftarrow R5 - R4$ |
| 4. ADD R6, R1, R8 | $R6 \leftarrow R1 + R8$ |
| 5. MUL R4, R9, R2 | $R4 \leftarrow R9 * R2$ |
| 6. SUB R9, R3, R4 | $R9 \leftarrow R3 - R4$ |

The number of RAW, WAW and WAR dependencies are respectively ?

- (A) 4, 2, 2 (B) 5, 2, 2
(C) 4, 1, 2 (D) 5, 1, 2

Q2 Consider the given set of instructions, which are having instruction format as follows:

Opcode, Destination operand, Source1, Source2

- | | |
|----------------|-------------------------|
| ADD R2, R1, R0 | $R2 \leftarrow R1 + R0$ |
| MUL R4, R3, R2 | $R4 \leftarrow R3 * R2$ |
| SUB R6, R5, R4 | $R6 \leftarrow R5 - R4$ |
| ADD R6, R7, R8 | $R6 \leftarrow R7 + R8$ |
| MUL R7, R1, R2 | $R7 \leftarrow R1 * R2$ |
| SUB R1, R3, R4 | $R1 \leftarrow R3 - R4$ |

The given instructions are executed in a 5 segment instruction pipeline which has segments as: Instruction Fetch, Instruction Decode, Operand Fetch, Execution and Write Back.

The speed up is calculated as follows :

Speed up =

$$\frac{\text{Number of cycles needed without operand forwarding}}{\text{Number of cycles needed with operand forwarding}}$$

Speed of the pipeline (correct up to 1 decimal place) for execution of above instructions is

_____?

Q3 The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB), The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 200 instructions. In the PO stage, 40 instructions take 4 clock cycles each, 65 instructions take 2 clock cycles each, and the remaining instructions take 1 clock cycle each. Assume that there are 20 instructions which cause 2 stalls each due to data hazards and there are no control hazards in program.

The number of clock cycles required for completion of execution of the sequence of instruction is _____?

Q4 Consider a 5 - stage instruction pipeline, where the stages take delays of 6 nanoseconds, 5 nanoseconds, 8 nanoseconds, 6 nanoseconds and 7 nanoseconds respectively. When an application is executing on this 5-stage pipeline, consider 20% of the instructions incur 3 pipeline stall cycles. The speed up (correct up to 2 decimal places) of the pipeline as compared to its corresponding non-pipeline system is _____?

Q5 Consider a non-pipelined processor operating at 10 GHz. It takes 4 clock cycles to complete an instruction. The same processor is upgraded to pipelined processor with 5 stages but clock rate



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is reduced to 8GHz. A program is executed on these processors which has 8% load/store instructions, 12% branch instructions and remaining ALU instructions. The memory access instructions cause 1 clock cycles each if there is no any cache miss but cause 40 cycles with cache miss. The 40% of all branch instructions cause 2 stalls each. There is no any stall associated with ALU instructions. Assume that cache has 94% hit. The speed up (round of up to 2 decimal place) achieved by pipeline over the non-pipeline processor for this program is _____?

- Q6** Which of the following statements is/are false regarding pipelining ?
- (A) For single input non-pipeline can perform better than equivalent pipeline system
 - (B) Pipeline can be fruitful when multiple different processing is applied over multiple inputs
 - (C) Pipeline with single segment cannot provide parallel processing
 - (D) Non-pipeline system does not require intermediate registers/buffer for

synchronization

- Q7** Consider a 6 segment pipeline in which all segments take 1 cycle for each instructions except the execution phase. Execution phase takes 3 cycle for multiply instruction, 6 cycles for division instruction and 1 cycle for addition and 1 cycle subtraction instruction. Suppose a code segment is executed on this pipeline in which total 8 instructions are executed and total 29 cycles needed for execution. Which of the following option(s) is/are correct regarding number of instructions of each type ?

(A)	ADD	SUB	MUL	DIV
	2	2	2	2
(B)	ADD	SUB	MUL	DIV
	1	1	3	3
(C)	ADD	SUB	MUL	DIV
	1	2	3	2
(D)	ADD	SUB	MUL	DIV
	2	1	2	3



Answer Key

Q1 (A)

Q2 1.4~1.4

Q3 429~429

Q4 2.5~2.5

Q5 2.35~2.35

Q6 (B)

Q7 (C)



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Hints & Solutions

Q1 Text Solution:

RAW dependencies are from instruction 1 to 2 for R2, from instruction 1 to 5 for R2, from instruction 2 to 3 for R4 and from instruction 5 to 6 for R4. There is no any RAW dependency for R4 from instruction 2 to 6 because R4 of instruction 6 depends on R4 of instruction 5 and not of R4 of instruction 2.

WAW dependencies from instruction 2 to 5 for R4 and from instruction 3 to 4 for R6.

WAR dependencies from instruction 3 to 5 for R4 and from instruction 5 to 6 for R9

Q2 Text Solution:

Solution: Number of data dependencies which can cause stall are: 2; which are from instruction 1 to 2 for R2, and from instruction 2 to 3 for R4.

Number of stalls due to each data dependency = stage number of write back - stage number of operand fetch

$$= 5 - 3$$

$$= 2$$

Number of stages (k) = 5

Number of instructions executed (n) = 6

Without operand forwarding total number of cycles = k + n - 1 + stall cycles for 2 data dependencies

$$= 5 + 6 - 1 + 2 * 2$$

$$= 14$$

With operand forwarding, the stalls due to dependencies are eliminated here, hence

With operand forwarding total number of cycles

$$= k + n - 1$$

$$= 5 + 6 - 1$$

$$= 10$$

$$\text{Hence speed up} = 14/10 = 1.4$$

Q3 Text Solution:

Stalls due to 40 instructions which take 4 cycles in PO stage = $40 * (4 - 1) = 120$

Stalls due to 65 instructions which take 2 cycles in PO stage = $65 * (2 - 1) = 65$

Stall due to data hazards = $20 * 2 = 40$

Total stall cycles = $120 + 65 + 40 = 225$

Number of stages (k) = 5

Number of instructions (n) = 200

Total number of cycles = k + n - 1 + total stall cycles

$$= 5 + 200 - 1 + 225$$

$$= 429$$

Q4 Text Solution:

One instruction execution time in non-pipeline system (tn) = $6 + 5 + 8 + 6 + 7 = 32\text{ns}$

CPI of pipeline system = $1 + 0.2 * 3 = 1.6$

Pipeline cycle time (tp) = max (6,5,8,6,7) = 8 nanoseconds

Speed up of pipeline = $32 / (1.6 * 8)$

$$= 2.5$$

Q5 Text Solution:

Cycle time in non-pipeline system = $1/10\text{GHz} = 0.1\text{nanoseconds}$

One instruction execution time in non-pipeline system = $4 * 0.1 = 0.4\text{ nanoseconds}$

Average Cycles per Instruction (CPI) in pipeline = $1 + 0.08 * 0.94 * 1 + 0.08 * 0.06 * 40 + 0.12 * 0.4 * 2$

$$= 1 + 0.2672 + 0.096$$

$$= 1.3632$$

Pipeline cycle time (tp) = $1/8\text{GHz} = 0.125\text{ nanoseconds}$

Speed up of pipeline = $0.4 / (1.3632 * 0.125) = 2.347 = 2.35$

Q6 Text Solution:



- (A) Option A is true because for 1 input pipeline time is $k \cdot t_p$, which is practically greater than one operation time in non-pipeline
- (B) Option B is false because pipeline is useful when same processing is to be applied over multiple inputs not different processing.
- (C) Option C is true because for parallel processing pipeline must have multiple segments or stages
- (D) Option D is true because intermediate buffer or register are used only in pipeline.

Q7 Text Solution:

Number of stall cycles due to structural hazard for each ADD instruction: $1 - 1 = 0$

Number of stall cycles due to structural hazard for each SUB instruction: $1 - 1 = 0$

Number of stall cycles due to structural hazard for each MUL instruction: $3 - 1 = 2$

Number of stall cycles due to structural hazard for each DIV instruction: $6 - 1 = 5$

Without stalls number of cycles = $k + n - 1 = 6 + 8 - 1 = 13$

For option A, number of stalls = $2 * 2 + 2 * 5 = 14$, hence total cycles = $13 + 14 = 27$

For option B, number of stalls = $3 * 2 + 3 * 5 = 21$, hence total cycles = $13 + 21 = 34$

For option C, number of stalls = $3 * 2 + 2 * 5 = 16$, hence total cycles = $13 + 16 = 29$

For option A, number of stalls = $2 * 2 + 23 * 5 = 19$, hence total cycles = $13 + 19 = 32$.


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